

3. Project Design Phase

Overall Design

The project is designed as a record-level security solution in ServiceNow. The custom table stores the required information, while ACLs decide whether a user can perform a particular operation on the records.

Table Design

The Institution Details table contains the fields required for maintaining student-related information. The Branch field is especially important because it provides the value used for the field-based READ restriction.

Role Design

The access model uses four roles. bb1 is used for READ access, bb2 is used for CREATE access, bb3 is used for WRITE access, and bb4 is used for DELETE access. Separating the roles makes it possible to control each operation individually.

ACL Design

A separate record ACL is created for each operation. The READ ACL uses the EEE branch condition and the advanced script. The CREATE, WRITE, and DELETE ACLs use their respective roles without a data condition.

Script Design

The READ script first checks whether the current user has the admin role. If the user is an administrator, access is allowed. It then checks for the bb1 role. If neither condition is satisfied, the script denies access.

Access Flow

When a user opens the table, ServiceNow evaluates the applicable ACL and the user's roles. For READ access, the Branch condition and advanced script are also considered. The final result is either permission to access the record or denial of access.